SPRINT RETROSPECTIVE – GLOF PROJECT

What went well	What went poorly	What ideas do you have	How should we take action
Successfully processed 66 Sentinel-2 datasets and generated clean temporal data	Cloud cover affected satellite data availability during some periods	Integrate radar-based data (Sentinel-1) to overcome cloud limitations	Add Sentinel-1 SAR data integration in next sprint
Automated DEM processing (sink filling, flow direction, watershed extraction) worked efficiently	DEM resolution (30m) caused minor inaccuracies in watershed boundaries	Use higher-resolution DEM or drone-based mapping	Explore UAV/drone-based data collection for critical zones
Decision Tree model provided accurate classification of glacier, lake, and terrain	Static NDWI thresholds initially caused false positives	Continue using ML models instead of static thresholds	Improve model training with more labeled datasets
Successfully extracted 413 Micro-Watersheds automatically	Manual validation of some watershed boundaries was still required	Automate validation using additional spatial checks	Add automated validation rules in pipeline
Morphometric parameters (AAR, SI, CCI) were computed correctly	Some parameters required heavy computation time	Optimize computation using parallel processing	Use optimized algorithms or cloud computing
GLOF Index successfully categorized lakes into risk levels	Very few high-risk lakes made validation difficult	Expand dataset to include more high-risk regions	Include more geographical regions in dataset
End-to-end pipeline automation reduced manual effort significantly	Pipeline depends heavily on optical data (affected by weather)	Combine multiple data sources (optical + radar + drone)	Implement hybrid data integration approach